

Working with Existing and New Environmental Data Management Systems

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Session 9

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11:00 – 12:30



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Environmental Data Management Systems Ecosystem

Meteorological, Hydrological, and Environmental organizations use a **wide variety of systems** to collect, store, manage, process, and share environmental observations, such as:



Aquarius • WIS2box • HydroServer • HYDATA • MIKE INFO • DataSight • Water Information Systems • Custom Databases

Hydrometeorological Data Systems in Participating Countries

These are some of the data management systems you mentioned using in the pilot worksheets you completed before the workshop.



Rwanda

AQUARIUS
Climsoft



Kenya

DataSight
Climsoft
Water Resources Information Management System
(WRIMS)



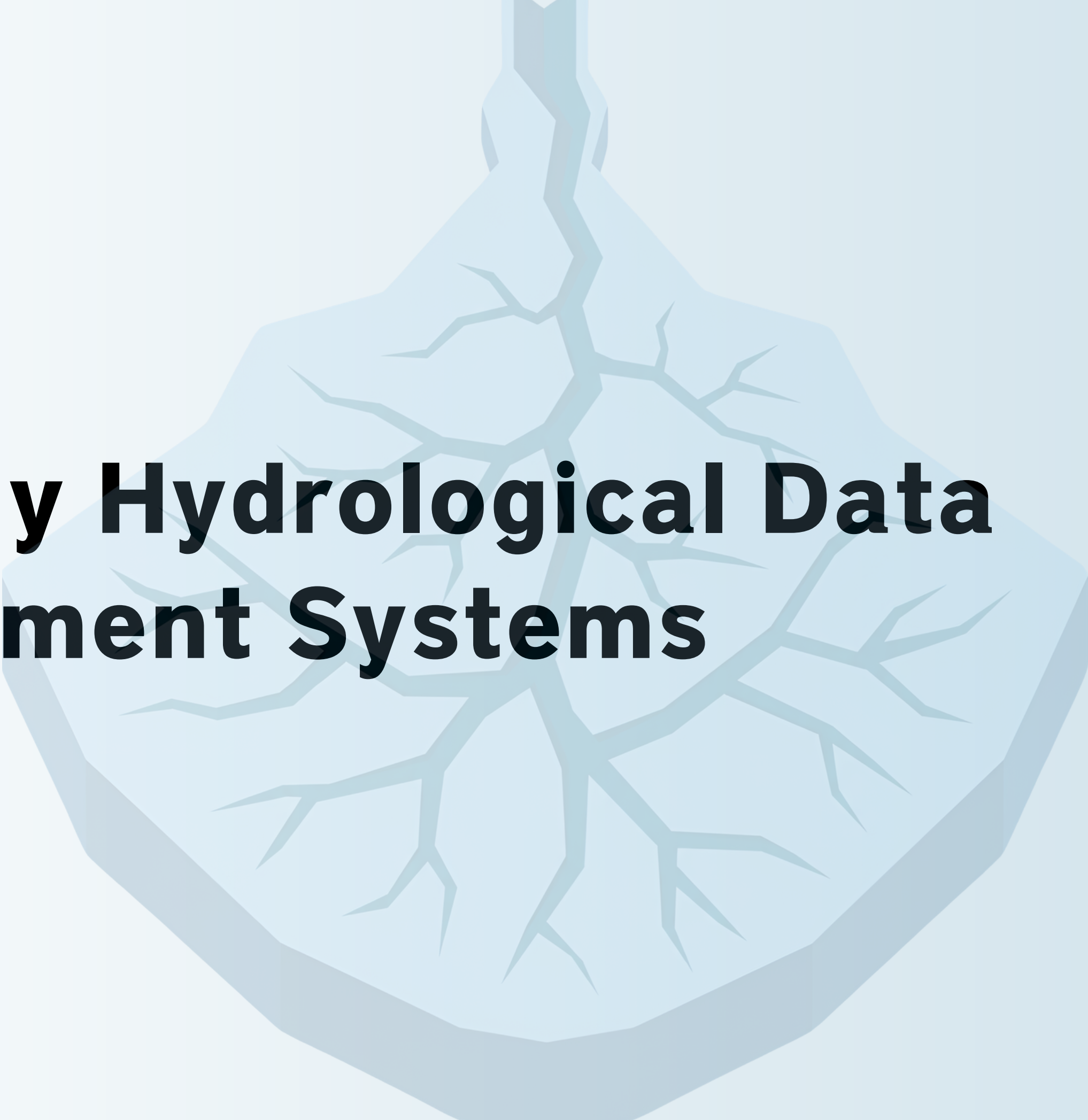
Ethiopia

HYDATA
AQUARIUS



Uganda

AQUARIUS
Climsoft
Water and Environment Information System (WEIS)



Primarily Hydrological Data Management Systems

- Commercial water and environmental data management platform from Aquatic Informatics.
- Manages real-time and historical monitoring data from multiple sources.
- Provides hydrology-specific QA/QC, data processing, and rating-curve tools.
- Supports visualization, analysis, reporting, and data publication.
- Cost: **Commercial license**/subscription; pricing on request.

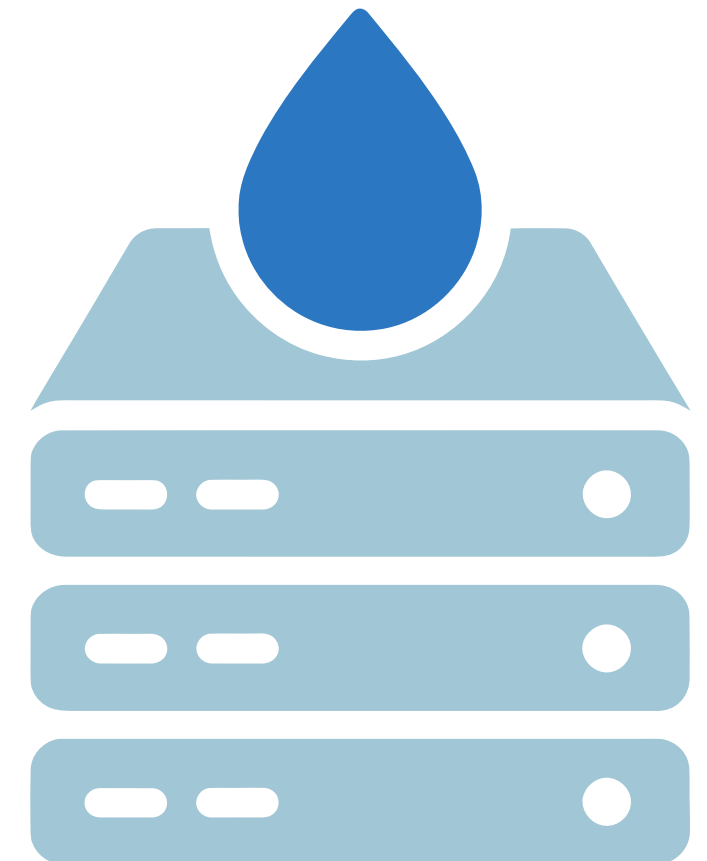


HYDATA

- **Legacy** hydrological database and analysis system.
- Manages water level, streamflow, rainfall, and reservoir data.
- Supports data import, validation, visualization, and export.
- Includes hydrology-specific rating curves, flow-duration curves, and low-flow analysis.
- Status: Legacy system; **no longer actively supported or maintained by the original developer.**
- Cost: Historically **commercial/licensed software**; no current pricing.



HYDATA 4.2



HYDROSERVER

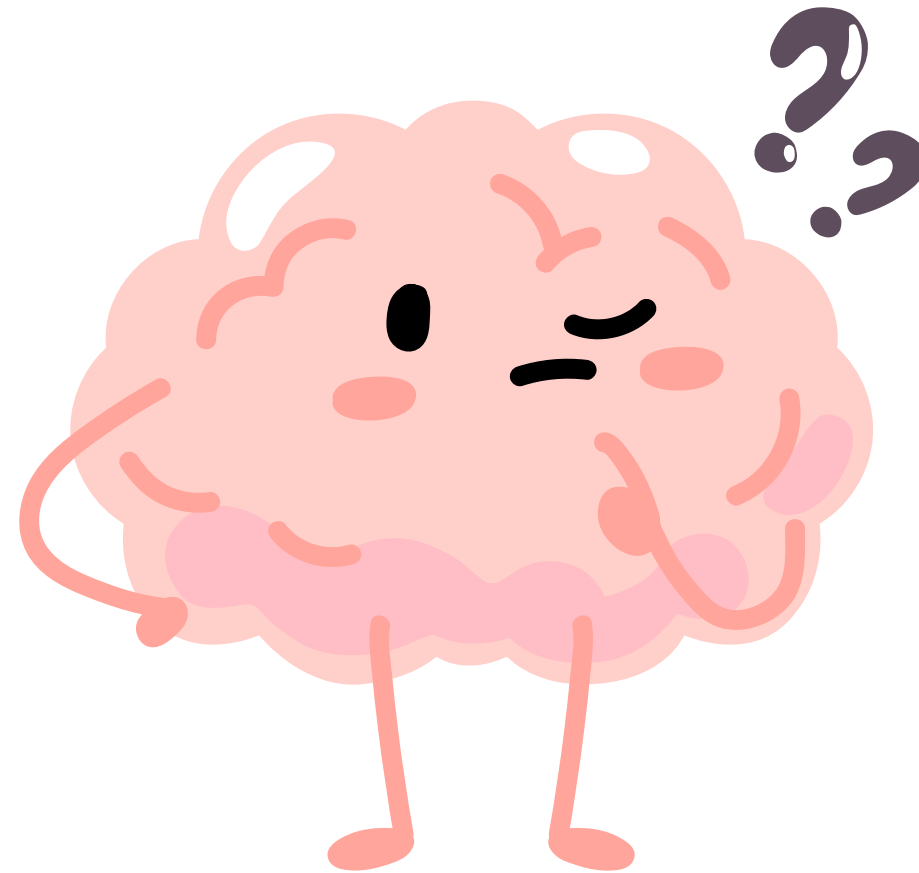
- Open-source platform for managing and sharing hydrologic and environmental data.
- Manages monitoring sites, time series, and associated metadata.
- Supports manual, programmatic, and automated data ingestion.
- Provides visualization, quality control, and API-based data access.
- Uses OGC SensorThings standards for **interoperability**.
- Status: **Actively developed and maintained**.
- Cost: **Free and open source**; hosting and maintenance may have costs.

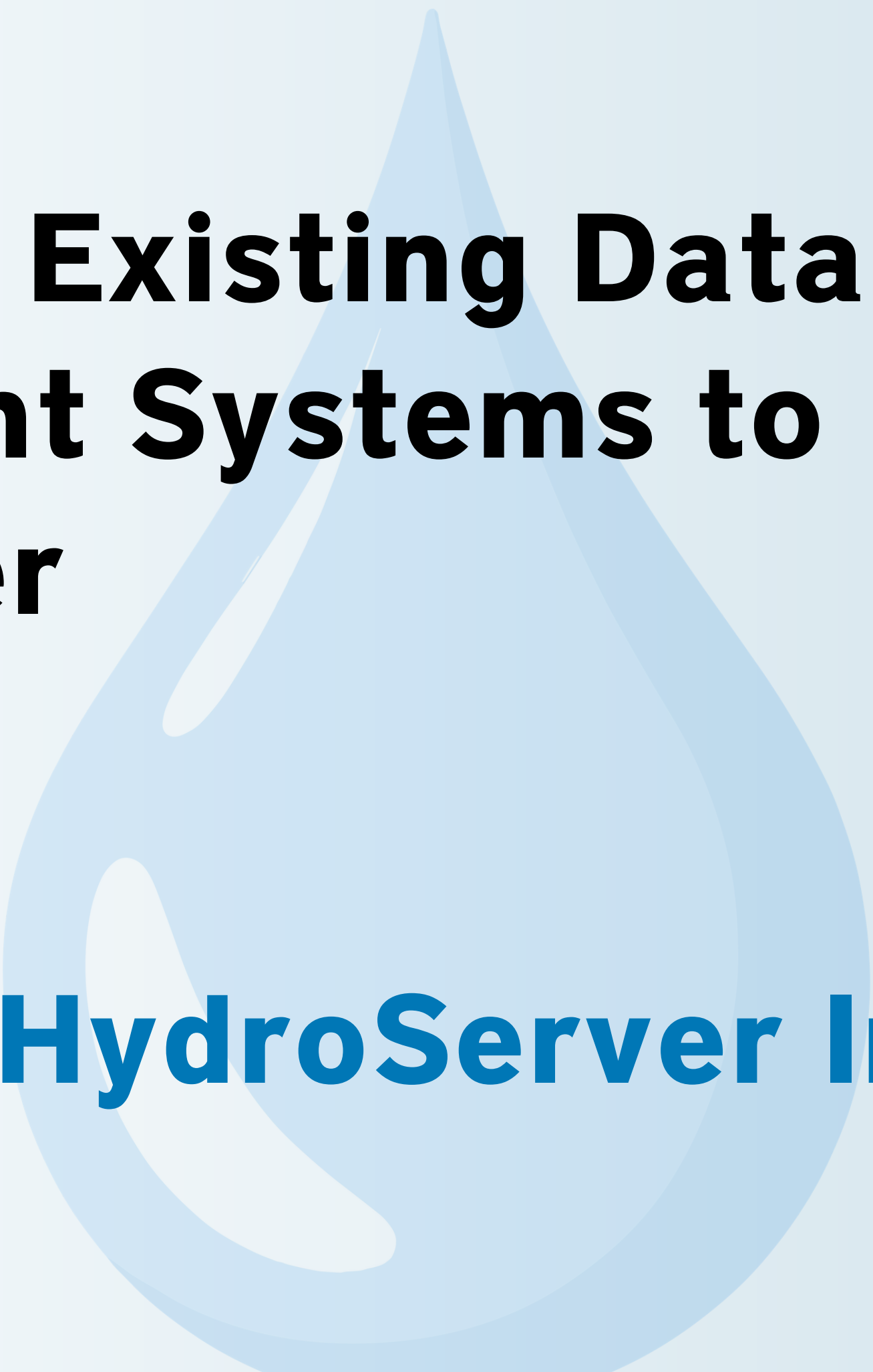


Comparison of Hydrological Data Management Systems

Capability	AQUARIUS	HYDATA	HydroServer
Historical time series	✓	✓	✓
Real-time data	✓	⚠	✓
Automated ingestion	✓	⚠	✓
Quality control	✓	✓	✓
Rating curves	✓	✓	✓
Web visualization	✓	✗	✓
API access	✓	✗	✓
OGC SensorThings-based	✗	✗	✓
Open source	✗	✗	✓
Actively developed	✓	✗	✓
Data privacy / access control	✓	⚠	✓

What if you already have a data management system but want to explore or start using a new one?





Connecting Existing Data Management Systems to HydroServer

Example:

AQUARIUS-HydroServer Integration

AQUARIUS-HydroServer Integration Workflow

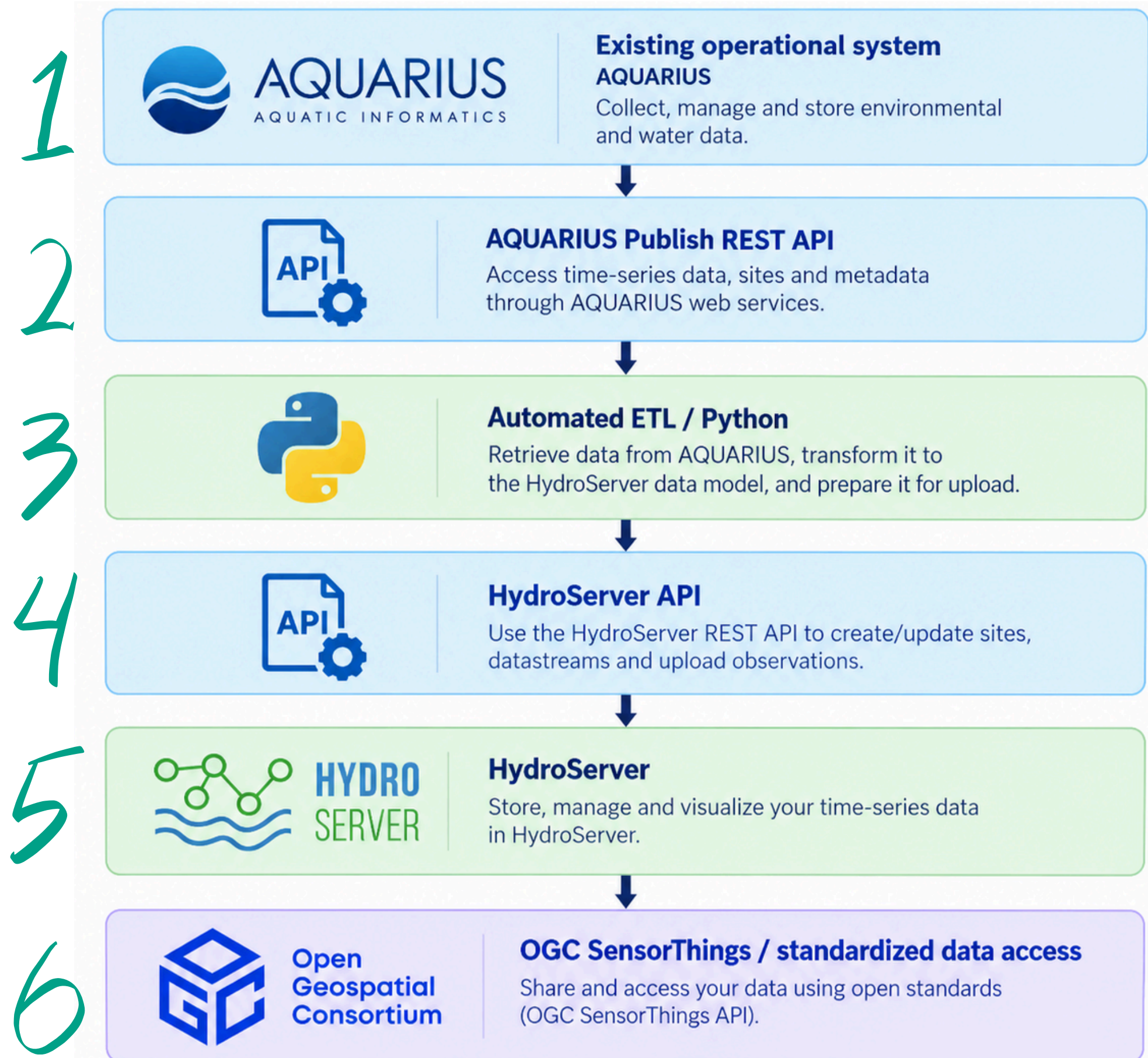
The workflow starts by retrieving data from AQUARIUS through the Publish REST API. The data are then extracted, transformed, and mapped using Python before being loaded into HydroServer through the HydroServer API.



NOTE: An API is a way for software to request or send data using defined web addresses. In this workflow, Python uses the AQUARIUS API to retrieve the data and the HydroServer API to load the data into HydroServer.

AQUARIUS-HydroServer Integration Steps

1. **AQUARIUS** - Existing data management system
2. **AQUARIUS API** - Access data and metadata
3. **Python** - Retrieve and transform data
4. **HydroServer API** - Load processed data
5. **HydroServer** - Store, manage, and visualize data
6. **OGC SensorThings** - Provide standardized data access



AQUARIUS-HydroServer Integration: A Practical Example

1. Identify the AQUARIUS time series.

Identify the location, time-series identifier, variable, unit, timezone, and other relevant metadata.

2. Connect to HydroServer.

Use the appropriate credentials or API authentication for HydroServer.

3. Create or identify the HydroServer datastream.

Create the required metadata and datastream, or identify an existing datastream where the observations will be loaded.

4. Retrieve the observations through the AQUARIUS Publish REST API.

Use Python to request the time-series data from AQUARIUS.

5. Transform the AQUARIUS response.

Convert timestamps and values to the format required by HydroServer.

6. Load Observations.

Send the transformed observations to the corresponding HydroServer datastream.

7. Verify it in Hydroservice and schedule as a task.

Verify the data in HydroServer, then schedule the ETL to run regularly and load new AQUARIUS observations as they become available.

We covered most of these steps during the HydroServer training, including connecting to HydroServer, creating resources such as monitoring sites and datastreams, retrieving time-series data from a web API, transforming data into the format required by HydroServer, loading observations, and scheduling tasks to upload new data regularly.

✓ All the steps covered during the HydroServer training

1. Identify the AQUARIUS time series.

Identify the location, time-series identifier, variable, unit, timezone, and other relevant metadata.

- ✓ 2. Connect to HydroServer.
- ✓ 3. Create or identify the HydroServer datastream.
- ✓ 4. Retrieve the observations through the AQUARIUS Publish REST API.
- ✓ 5. Transform the AQUARIUS response.
- ✓ 6. Load Observations
- ✓ 7. Verify it in Hydroserver and scehdule as a task

We learned how to retrieve and transform data using the GEOGLOWS API in Exercises 4 and 5.

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To upload existing **historical observations** from AQUARIUS to HydroServer (without regular updates), refer to Exercise 1 of the HydroServer training.

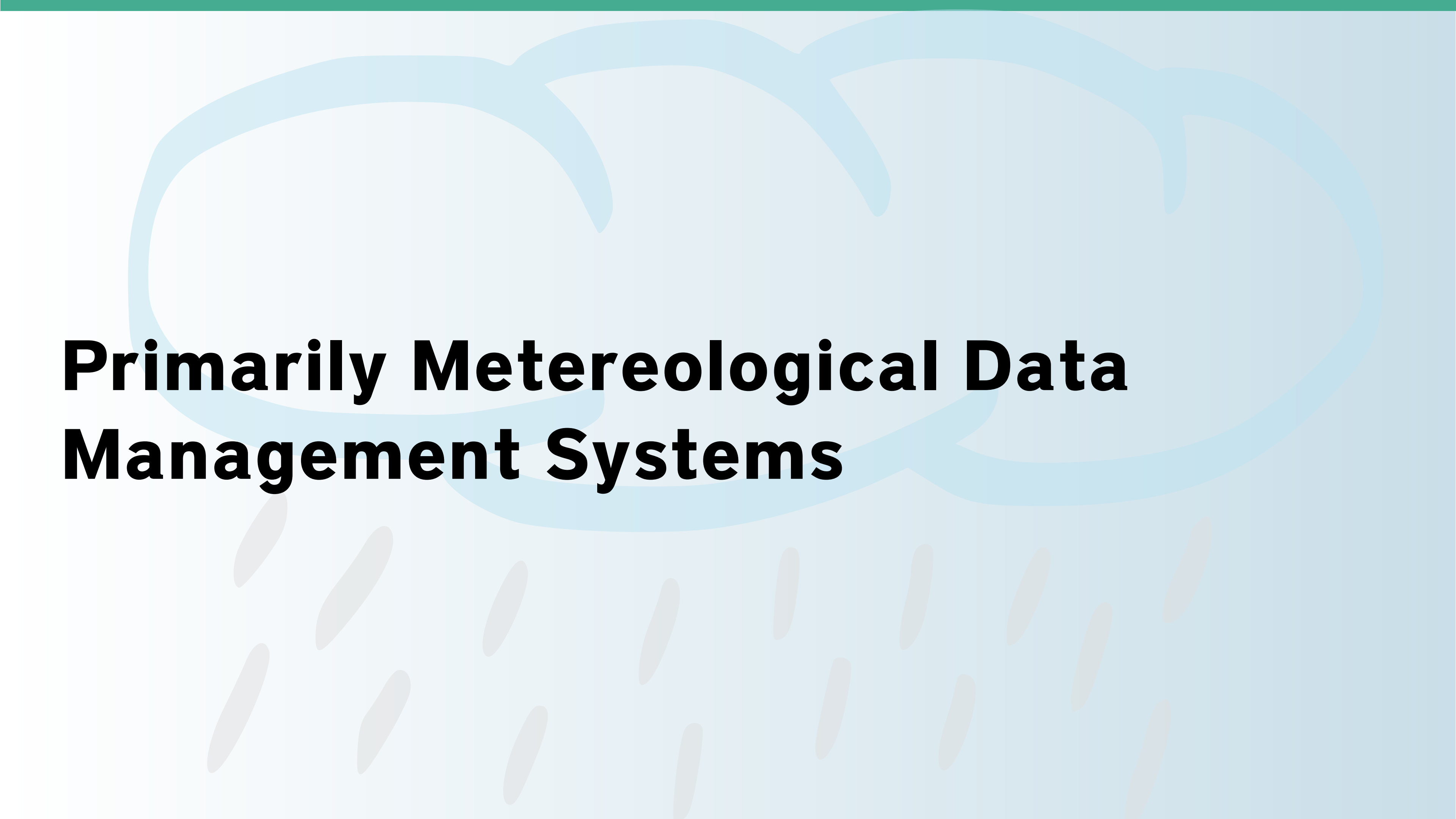
To **automate the ingestion** of real-time observations from AQUARIUS, set up a scheduled ETL task using either an externally managed ETL workflow (**Exercise 4**) or a HydroServer-managed ETL workflow (**Exercise 5**).



You can use this workflow not only for hydrological data, such as stage and discharge, but also for meteorological and other environmental data stored in AQUARIUS.

As discussed during the HydroServer training, HydroServer is not limited to hydrological data.

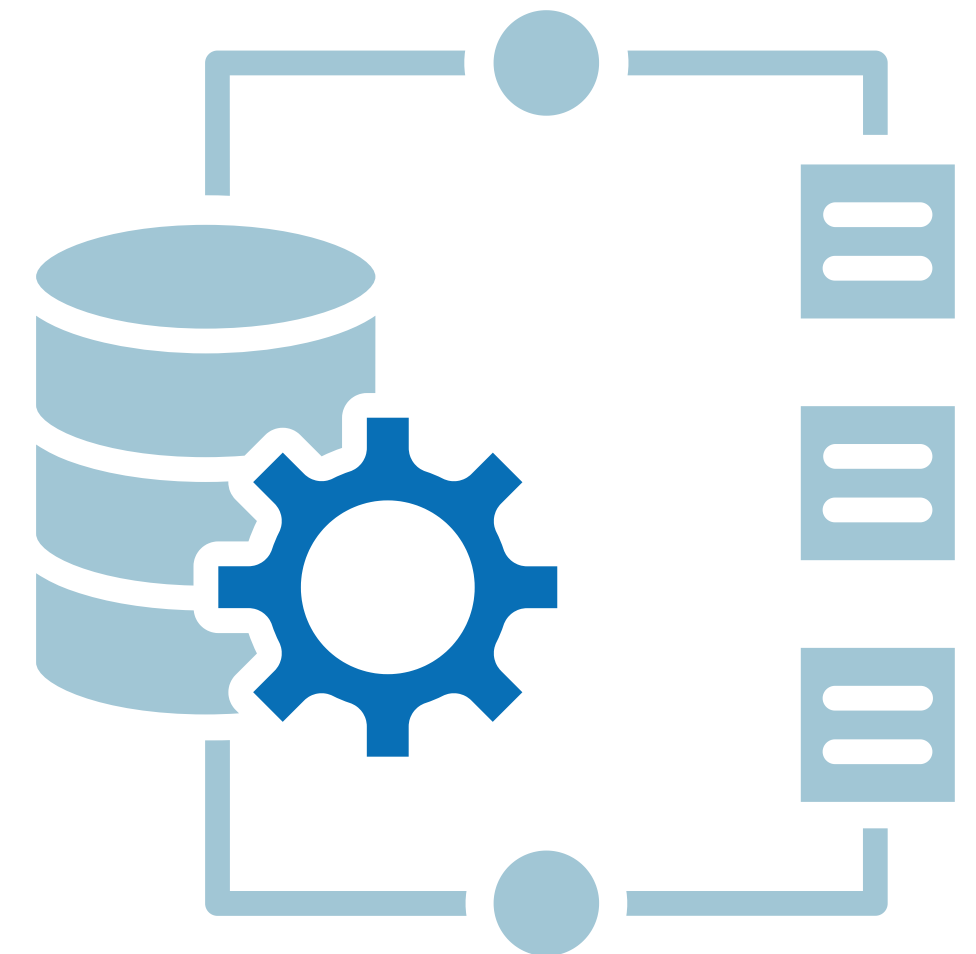
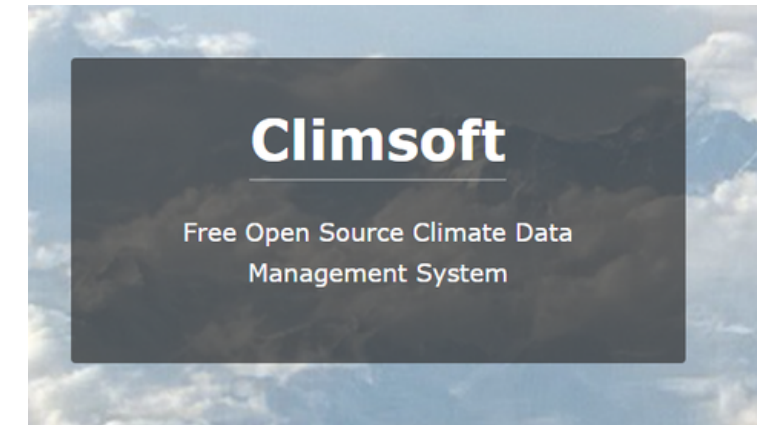




Primarily Meteorological Data Management Systems

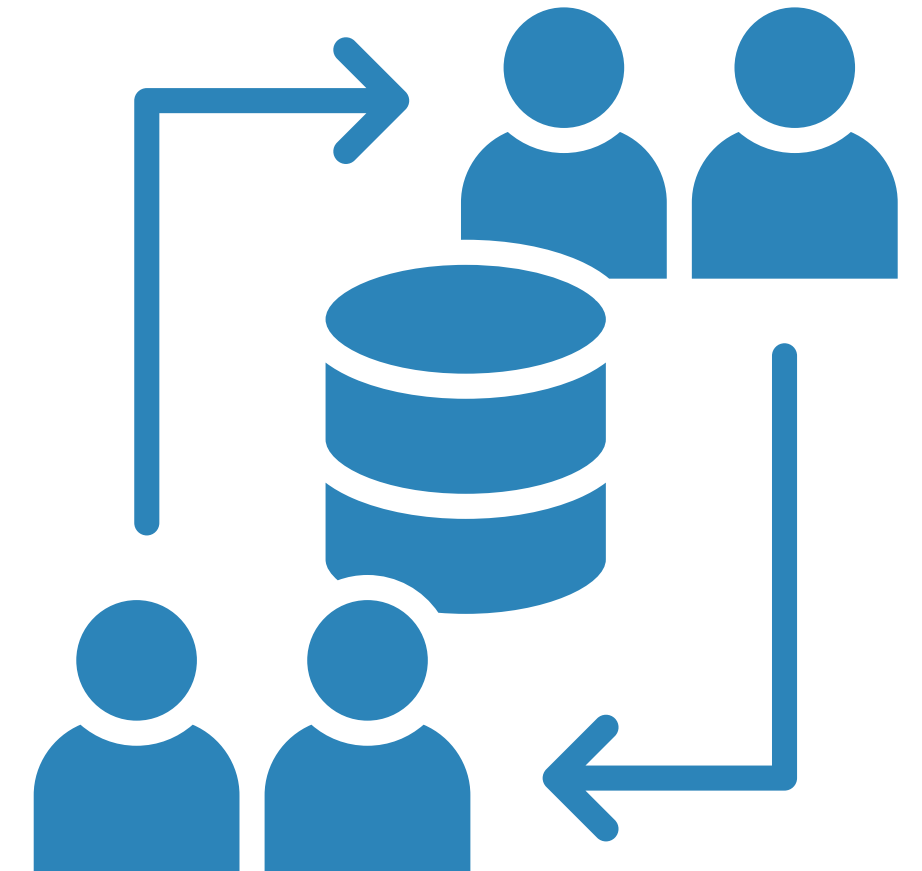
Climsoft

- Free, open-source Climate Data Management System (CDMS) for **meteorological services**.
- Manages data from manual and automatic weather stations.
- Supports data ingestion, quality control, storage, retrieval, and reporting.
- **Supports WMO standards** and meteorological data formats.
- Cost: **Free and open source**; deployment and maintenance may have costs.



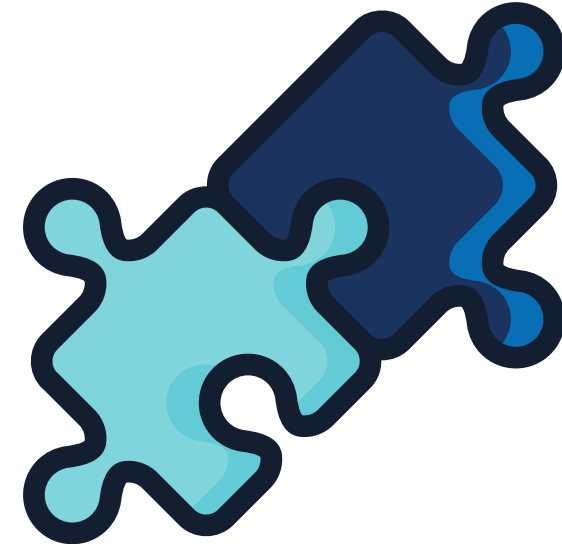
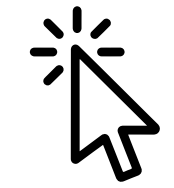
WIS2BOX

- Free, open-source software developed to help WMO Members connect to WIS 2.0 and share their data.
- Enables weather, climate, and water data to be published and exchanged through WIS 2.0.
- Supports data ingestion, transformation, publication, discovery, and visualization.
- Supports WMO standards, including WMO data formats, OGC APIs, and MQTT-based notifications.
- Can be deployed on-premises or in the cloud using Docker.
- Cost: **Free and open source**; infrastructure, deployment, and maintenance may have costs.



**Push CLIMSOFT data to WIS2 using
WIS2BOX**

Guidance on how to establish the connection between Climsoft and WIS2BOX is available in the WIS2 in a Box documentation.



You can also follow the step-by-step instructions in Slides 21–25 to transfer data from Climsoft to WIS2BOX in just four steps.

Climsoft Web

≡ Export Specifications

Dashboard

Data Ingestion

Data Monitoring

Quality Control

Data Extraction

Metadata

- Elements
- Flags
- Organisations
- Network Affiliations
- Regions
- Stations
- QC Specifications
- Source Specifications
- Export Specifications**
- Connector Specifications
- Adapter Specifications

System Administrator

Add

Options

Name	Type	Description	Actions
No exports yet. Click "Add" to create one.			

In Climsoft Web, go to Export Specifications, then click Add.

Define how Climsoft database elements are mapped to the corresponding WIS2BOX elements.

Climsoft Web

Export Specifications

Dashboard

Data Ingestion

Data Monitoring

Quality Control

Data Extraction

Metadata

→ Elements

→ Flags

→ Organisations

→ Network Affiliations

→ Regions

→ Stations

→ QC Specifications

→ Source Specifications

→ Export Specifications

→ Connector Specifications

→ Adapter Specifications

System Administrator

New Export Specification

Name:

Description:

Export Type:

Dissemination

WIS2BOX Configuration

Report Type:

SYNOPSIS

Element Mappings

Add Mapping

Database Element	WIS2BOX Element

Post-Export Adapter (optional):

No adapter

Submit Cancel

Set the Export Type to "Dissemination."

Select SYNOPSIS as the Report Type.

Click Add Mapping, then map each Climsoft Database Element to its corresponding WIS2BOX Element.

Climsoft Web

≡ Connector Specifications

Dashboard

Data Ingestion

Data Monitoring

Quality Control

Data Extraction

Metadata

- Elements
- Flags
- Organisations
- Network Affiliations
- Regions
- Stations
- QC Specifications
- Source Specifications
- Export Specifications
- Connector Specifications
- Adapter Specifications

System Administrator

AddOptions

Name	Type	Description	Host	Schedule	Actions
No connectors yet. Click "Add" to create one.					

Then, go to Connector Specifications and click Add.

Configure where and when the exported data will be sent to WIS2BOX.

Climsoft Web **Connector Specifications**

Name

Description

Connector Type:
EXPORT

Server Type:
FILE SERVER

Host Name ip address or domain.

Timeout Seconds.
300

Retry Attempts retries after the first attempt; 0 = no retries.
0

Schedule cron expression.
e.g. */5 * * * *

Submit **Cancel**

Enter the required information, such as the name and description.

Set the Connector Type to "Export" and the Server Type to "File Server."

Configure where and when the exported data will be sent to WIS2BOX.

New Connector Specification

Remote Path: /

Observation Window

Each scheduled run exports observations whose chosen date falls within the look-back window.

Look back the last^{minutes}: 1450

Filter by: Observation date

Export Specifications [Add Specification](#)

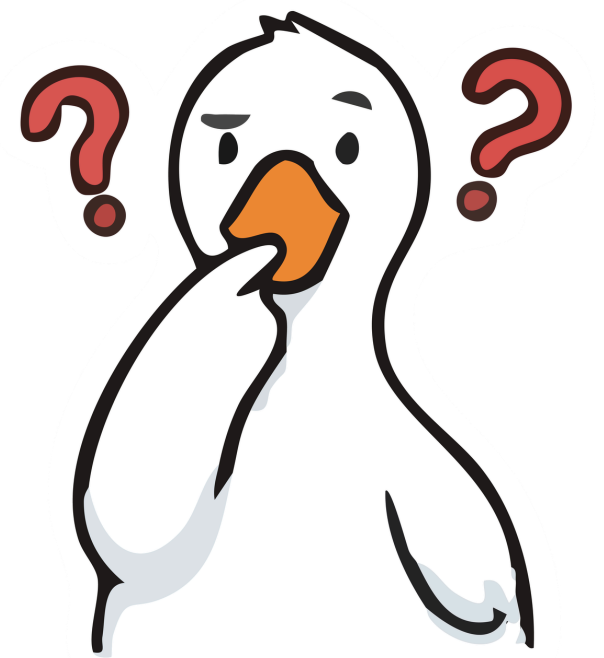
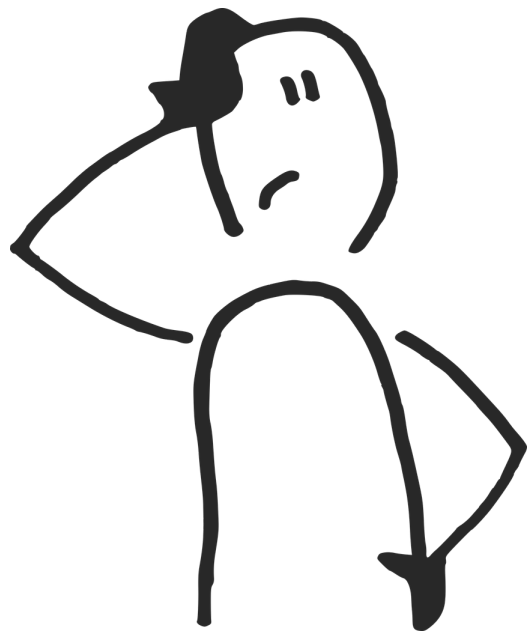
Export Specification	Station

Comment:

[Submit](#) [Cancel](#)

**Let's Take some time to Discuss and
Share Our Experiences**

Any Question?

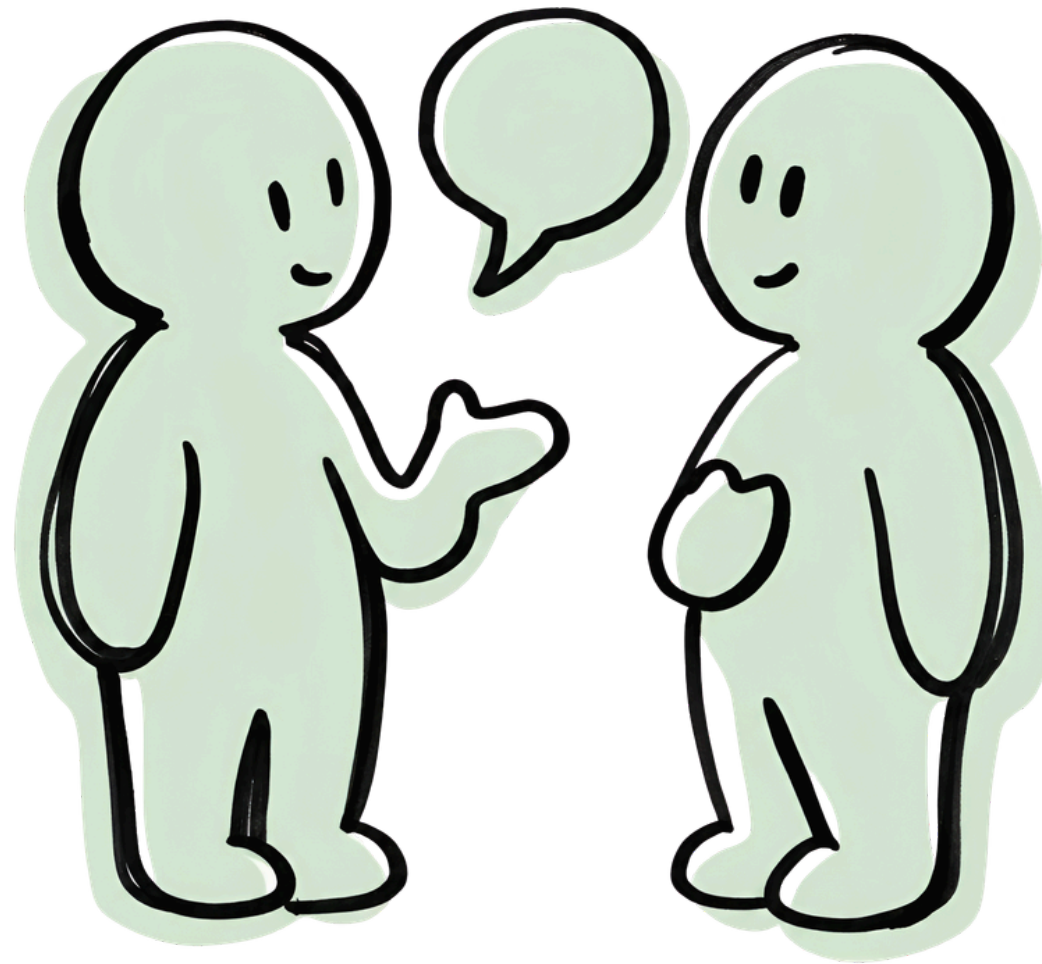


Which data sources are you envisioning connecting to HydroServer and WIS2BOX?

Let's share our experiences/questions and learn from each other!

Do you have previous experience connecting different data management systems?

What challenges do you anticipate when integrating your existing systems with HydroServer and WIS2BOX?



Which parts of the integration process do you think will require the most support?

See you in the next session!

Now, take a break!

